STULZ

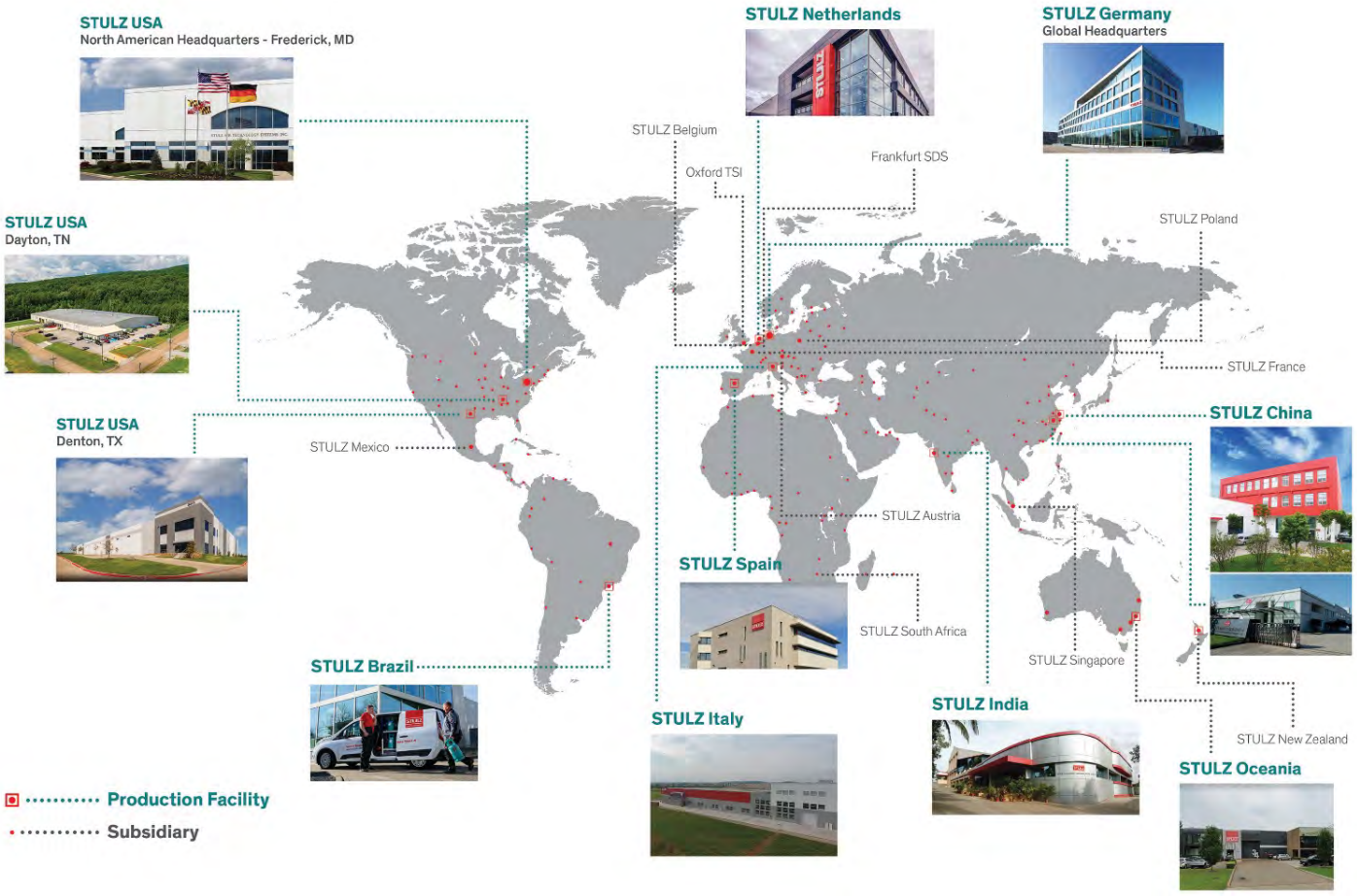
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CyberRow DX

Engineering Manual

Row-Based Precision Air Conditioners
12 kW - 33 kW / 60 Hz



ABOUT STULZ

Since it was founded in 1947, the STULZ company has evolved into one of the world's leading system suppliers of computer room air conditioners and air handlers.

STULZ has seen continual international expansion of its precision cooling business, specializing in temperature and humidity control for data centers and telecommunications installations.

STULZ has ten production plants (three in the USA, one each in Germany, China, Italy, Great Britain, Spain, Brazil, and India), and twenty-eight subsidiaries (in Germany, USA, France, Italy, Great Britain, Belgium, Brazil, the Netherlands, New Zealand, Mexico, Austria, Poland, Spain, Singapore, China, India, Indonesia, South Africa, Sweden, and Australia).

STULZ works with sales and service partners in over 140 other countries, and therefore boasts an international network of cooling specialists.

The benchmark of flexibility and energy efficiency

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Nomenclature

CRS-XXX-X		
System	**Nominal Capacity (MBH)**	**Cooling Method**
CyberRow Row Based Cooling	**1000s of Btu/h (MBH)**	**AR = Remote (Split Air Cooled) G = Glycol Cooled W = Water Cooled**

CRS -084-W

CyberRow System

Nominal Capacity in MBH (1,000s of Btu/h)

Water Cooled System



Model Nomenclature Specifications

Air Cooled Remote Evaporator (Models CRS-___-AR)

The floor mounted precision air conditioner system shall be a split air cooled evaporator with remote air cooled condenser. The evaporator section shall house, at a minimum, the evaporator coil, expansion valve, compressor, evaporator blower/motor and associated electrical and refrigeration components.



Air Cooled Heat Rejection

The remote outdoor air-cooled condenser shall be sized to provide the total heat of rejection of the system and incorporate low ambient head pressure control method, -20°F or -30°F.

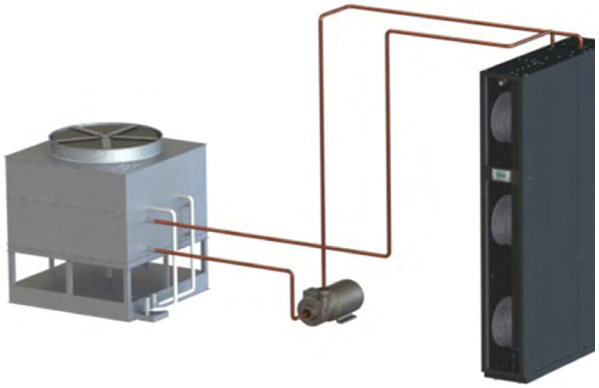
DX Water Cooled (Models CRS-___-W)

The floor mounted precision air conditioner system shall be self-contained to include an integral water cooled, plate-fin condenser with factory installed head pressure water regulating control valve(s).

Condenser(source) water shall be provided by a cooling tower or some other remote water source.

Head Pressure Control

Head pressure shall be automatically controlled by factory installed two-way, or three-way water regulating valves rated for 600 psig w.w.p.

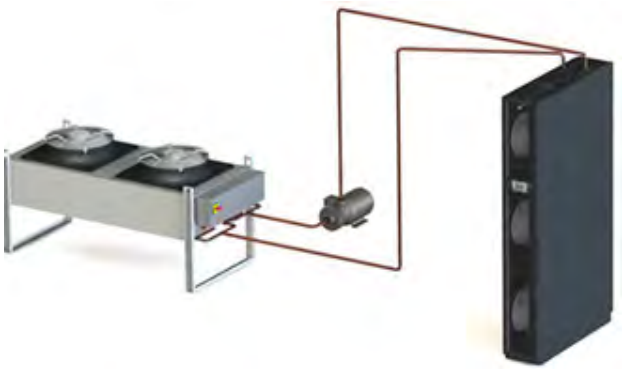


Water Cooled Head Pressure Control

Head pressure shall be automatically controlled by factory installed two-way, or three-way water regulating valves rated for 600 psig w.w.p.

DX Glycol Cooled (Models CRS-___-G)

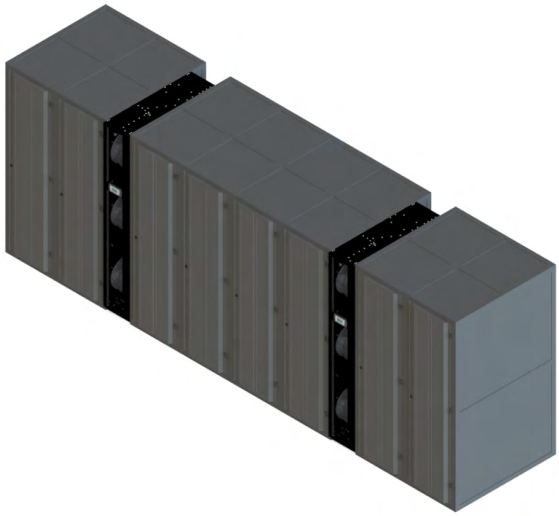
The floor mounted precision air conditioner system shall be self-contained to include an integral glycol cooled, plate-fin condenser with factory installed head pressure glycol regulating control valve. Condenser (source) glycol solution shall be provided via a STULZ model GPS - _ -_ remote glycol pump package and F _ S -_ dry cooler system.



Glycol Regulating Valve

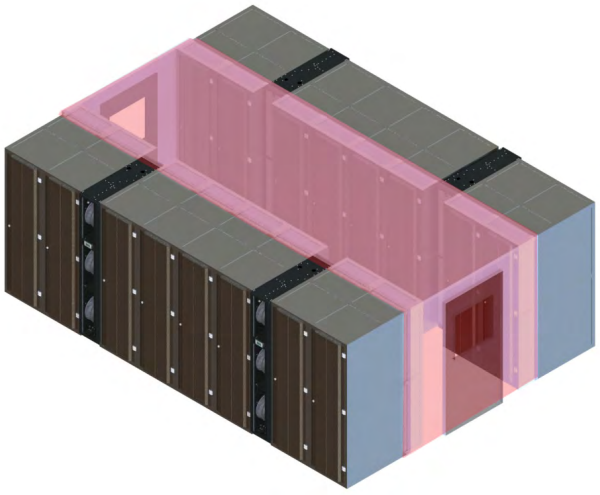
Head pressure shall be automatically controlled by factory installed two-way, or three-way, water regulating valve rated for 600 psig w.w.p.

Configurations



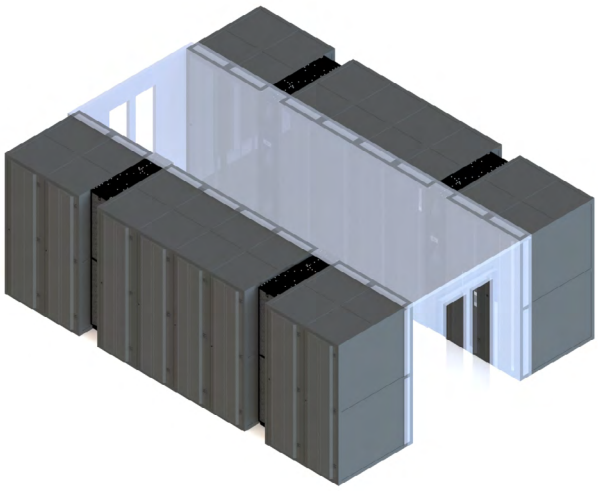
Open Aisle

Open aisle configuration organizes racks in a single row or in hot and cold aisle rows, but without containment. The CyberRow draws hot air from the external environment or hot aisle, removes the heat, and supplies cooled air to the front of IT equipment in the cold aisle.



Hot Aisle

Hot aisle containment captures the hot exhaust air from IT equipment and prevents from mixing with cool air. The front of IT equipment is accessed in the external cold aisle. The CyberRow draws the contained hot air from the hot aisle, removes the heat, and supplies cooled air into the external cold aisle.



Cold Aisle

Cold aisle containment captures cooled air from the CyberRow and prevents it from mixing with hot air. The front of IT equipment is accessed in the contained cold aisle. The CyberRow draws hot air from the external environment, removes the heat, and supplies cooled air back into the contained cold aisle.

Table continued on page 18

Standard Product Features

Models CRS-042 through CRS-084

Model: CRS-	042	084
SELECTED FEATURES		
Cabinet		
Galvannealed Steel, Black Powder Coated Finish	Standard	Standard
Air Pattern and Filtration		
Front Discharge	Standard	Standard
Front Diverted Plenum Discharge	Optional	Optional
Permanent Washable Filters	Standard	Standard
Mechanical Components		
Backward Inclined, Plenum Style Fan, with an EC Motor	Standard	Standard
R-410A Refrigerant	Standard	Standard
Scroll Compressor	Standard	Standard
Variable Compressor Speed Control	N/A	N/A
Proportional Electronic Expansion Valve	Standard	Standard
Electronic Hot Gas Bypass Valve	Standard	Standard
Piping Configuration (Top or Bottom)	Standard	Standard
Condensate Pump	Optional	Optional
Electrical System		
Voltage and Power Supply	Standard	Standard
Single Point Power Connection	Standard	Standard
Dual Power Connection	N/A	N/A
Remote Stop/Start Contacts	Standard	Standard
Main Power Switch	Standard	Standard
Humidity Control		
5 lb Electrode Canister Steam Humidifier	N/A	N/A
9 kW Electric Heat/Reheat	N/A	N/A
E² Microprocessor Controller		
A/C Grouping pLAN Interface	Optional	Optional
BMS Interface	Optional	Optional
Common Alarm, Dry Contact	Standard	Standard
Optional Accessories		
Smoke Detection	Optional	Optional
Code Conformance		
Certified and listed to UL 60335-1 and UL 60335-2-40	Standard	Standard
Certified to CSA C22.2 # 60335-1 and CSA C22.2 # 60335-2-40	Standard	Standard

*Note: Piping connection location to be specified at time of order.

Standard Product Features - Continued

Models CRS-042 through CRS-084

Model: CRS-	042	084
SELECTED FEATURES		
Specific Model Standard Features		
Remote Air Cooled Condenser Rated for 95°F or 105°F High Ambient	Standard	Standard
Head Pressure Control (-20°F Low Ambient Variable Fan Speed Control)	Standard	Standard
Head Pressure Control (-30°F Low Ambient Flooded and Fan Speed Control)	Optional	Optional
Water / Glycol Cooled Systems		
Stainless Steel Brazed-Plate (W/G)	Standard	Standard
Two-Way 600 psig Regulating Valve (W/G)	Standard	Standard
Three-Way 600 psig Regulating Valve (W/G)	Optional	Optional
Free Cooling / Alternate Water Source Systems		
Three-Way, 600 psig Water / Glycol Control Valve	N/A	N/A

Technical Data

Air Cooled Systems - Models 042 - 084

Model: CRS-	042	084
Blower Motor: Backward Inclined, Plenum Style Fan, with an EC Motor		
Horsepower per Motor, hp	1/4	1/4
Air Flow Rate, CFM	1,500	2,900
Quantity of Blowers	3	3
Drive Method	Direct	Direct
Direct Expansion (DX) Coil		
Evaporator Coil: Aluminum Fin, Copper Tube		
Rows (Face Area, ft^2)	3 (5.8)	4 (5.8)
Face Velocity, ft/min	260	500
Compressor, Heat Pump Duty Rated Scroll, R-410A		
Type (Qty)	Scroll (1)	Scroll (1)
Watts Input	3,680	7,090
Total Heat of Rejection, MBH	64.6	124
Condenser Type	Remote Air Cooled	Remote Air Cooled
Head Pressure Control	See Remote Condenser Selection	
Connection Sizes, Copper		
Condensate Drain with Pump, in FPT	1/2	1/2
Hot Gas & Liquid Refrigerant Connections (Air Cooled), in O.D.	5/8	5/8
Physical Data		
Approximate Weight, lb	450	480
Dimensions: (H × W × D), in without Side Diverter Panel Option	77.8 × 11.6 × 42.1	77.8 × 11.6 × 42.1
Dimensions: (H × W × D), in with Side Diverter Panel Option	77.8 × 11.6 × 48.3	77.8 × 11.6 × 48.3
Approximate Shipping Weight, lb	636	665
Approximate Shipping Dimensions: (H × W × D), in	84 × 22 × 54	84 × 22 × 54

Technical Data

Water, Glycol Cooled Systems - Models 042 - 084

Model: CRS-	042		084	
	-W	-G	-W	-G
Blower Motor: Backward Inclined, Plenum Style Fan, with an EC Motor				
Horsepower per Motor, hp	1/4	1/4	1/4	1/4
Air Flow Rate, CFM	1,500	1,500	2,900	2,900
Quantity of Blowers	3	3	3	3
Drive Method	Direct	Direct	Direct	Direct
Direct Expansion (DX) Coil				
Evaporator Coil: Aluminum Fin, Copper Tube				
Rows (Face Area, ft^2)	3 (5.8)	3 (5.8)	4 (5.8)	4 (5.8)
Face Velocity, ft/min	260	260	500	500
Economizer Coil: Aluminum Fin, Copper Tube				
Rows (Face Area, ft^2)	N/A	N/A	N/A	N/A
Face Velocity, ft/min	N/A	N/A	N/A	N/A
Head Pressure Control				
Standard Control	Modulating Two-Way Water Valve			
Optional Control	Modulating Three-Way Water Valve			
Compressor: Heat Pump Duty Rated Scroll, R-410A				
Type (Qty)	Scroll (1)	Scroll (1)	Scroll (1)	Scroll (1)
Watts Input	2,820	3,940	5,690	7,470
Total Heat of Rejection, MBH	66	64	128	124
Flow Rate, GPM @ 85°F EWT / 95°F LWT 0% Glycol Solution	13.2	N/A	25.6	N/A
Flow Rate, GPM @ 110°F EGT / 120°F LGT 40% Glycol Solution	N/A	13.9	N/A	26.7
Unit Pressure Drop, ft wg	28.6	34.4	21.5	25.2
Condenser Type	Integral Brazed Plate			
Connection Sizes Copper				
Condensate Drain with Pump, in FPT	1/2	1/2	1/2	1/2
Condenser Fluid In / Out, in NPT	1 1/4	1 1/4	1 1/4	1 1/4
Physical Data				
Approximate Weight, lb	520	520	550	550
Dimensions: (H × W × D), in without Side Diverter Panel Option	77.8 × 11.6 × 42.1			
Dimensions: (H × W × D), in with Side Diverter Panel Option	77.8 × 11.6 × 48.3			
Approximate Shipping Weight, lb	706	706	736	736
Approximate Shipping Dimensions: (H × W × D), in	84 × 22 × 54			

Performance Data

Air , Water, Glycol Cooled Systems - Models 042 and 084

Model: CRS-		042 - AR	042 - W	042-G	084-AR	084-W	084-G
Net DX Cooling Capacity - MBH / kW (Includes Motor heat @ Rated CFM)							
100°FDB / 69.2°FWB Entering Air Temperature							
Total	MBH	52.0	55.0	51.0	99.0	107.0	97.0
Sensible		52.0	55.0	51.0	99.0	107.0	97.0
Total	kW	15.1	16.2	14.8	28.9	31.3	28.3
Sensible		15.1	16.2	14.8	28.9	31.0	28.3
95°FDB / 67.7°FWB Entering Air Temperature							
Total	MBH	49.0	53.0	48.0	94.0	101.0	92.0
Sensible		49.0	53.0	48.0	94.0	101.0	92.0
Total	kW	14.3	15.4	14.0	27.4	29.7	26.8
Sensible		14.3	15.4	14.0	27.4	29.7	26.8
90°FDB / 66.1°FWB Entering Air Temperature							
Total	MBH	46.0	50.0	45.0	88.0	97.0	88.0
Sensible		46.0	50.0	45.0	88.0	97.0	88.0
Total	kW	13.5	14.7	13.2	25.8	28.5	25.7
Sensible		13.5	14.7	13.2	25.8	28.5	25.7
85°FDB / 64.5°FWB Entering Air Temperature							
Total	MBH	44,0	49.0	43.0	85.0	95.0	82.p
Sensible		44.0	49.0	43.0	85.0	95.0	82.0
Total	kW	12.8	14.2	12.5	24.9	27.8	23.9
Sensible		12.8	14.1	12.5	24.9	27.8	23.9
80°FDB / 62.8°FWB Entering Air Temperature							
Total	MBH	42.0	47.0	41.0	83.0	92.0	81.0
Sensible		42.0	47.0	41.0	83.0	88.0	81.0
Total	kW	12.4	13.7	12.1	24.2	26.8	23.7
Sensible		12.4	13.7	12.0	24.1	25.8	23.7

Electrical Data

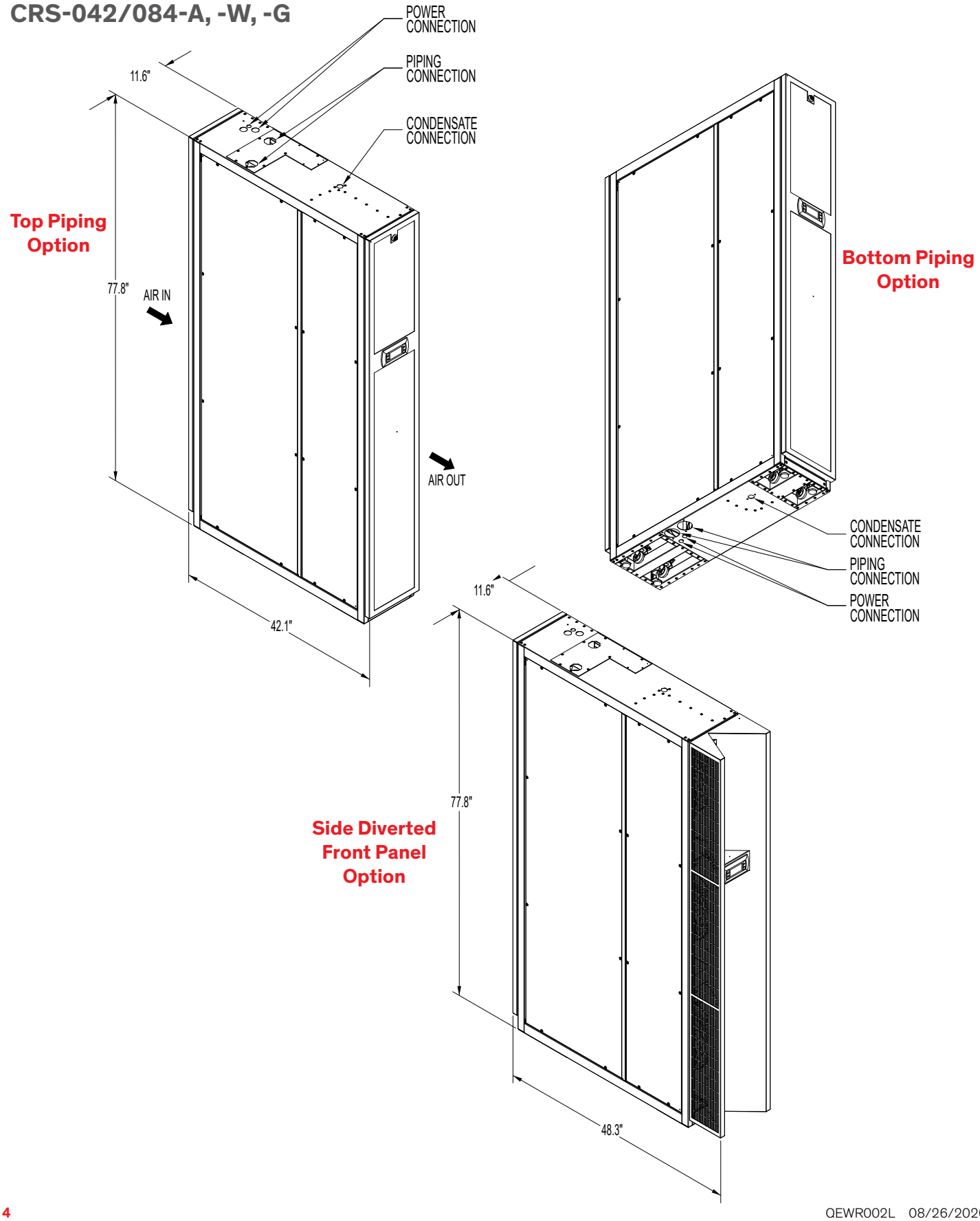
Models 042 - 084

Model: CRS-	042		084	
Voltages	MCA	MFS	MCA	MFS
Cooling Only (Includes Condensate Pump)				
208 V / 1 PH / 60 Hz	31.9	50	39.3	60
208 V / 3 PH / 60 Hz	25.6	40	38.1	60
460 V / 3 PH / 60 Hz	12.2	15	18.8	30
575 V / 3 PH / 60 Hz	10	15	13.7	20

Note: Standard 1 KAIC rating, consult factory for alternative KAIC ratings

Dimensional Drawings

CRS-042/084-A, -W, -G



Guide Specifications

CyberRow DX

Summary

Row-Based Precision Control Air Conditioners Direct Expansion — Air, Water, Glycol

This specification describes requirements for a precision environmental control system. The STULZ CyberRow is a row- based cooling system that shall provide precision temperature control for computer rooms, or rooms containing telecommunications or other highly sensitive heat load equipment, where continuous 24 hours a day 365 days a year air conditioning is required. Designed with both front and rear access, CyberRow systems require minimum floor space. The supplied system shall be provided with ETL Certification. The CyberRow model number shall be CRS - _ _ _ - _ .

Design Requirements
The environmental control system shall be a CyberRow Direct Expansion (DX) factory-assembled unit. The unit shall be designed for a row-based installation with removable front and rear access panels. No allowance for side service access shall be required, however removable side access panels shall be provided for additional access. CyberRow units are especially adapted for both raised and non- raised floors. The air handling system shall be specifically designed for high sensible heat ratio. Each system shall be capable of handling CFM. The unit shall have the cooling capacity in Btu/h, and the sensible cooling capacity in Btu/h based on entering air conditions in °F dry bulb (DB) and °F wet bulb (WB). The main fan motors shall be in HP. The unit shall have a power supply of in volts. See the performance data section for details, by model.

Quality Assurance
The manufacturer shall maintain a set of international standards of quality management to ensure product quality. Each system shall be subjected to a complete operational and functional test procedure at the factory prior to shipment.

Cabinet
Side access panels shall be fabricated from 20 gauge galvannealed steel and shall be securely bolted to a 14 gauge base and the top plate. The top plate, front and rear panels shall be fabricated from 16 gauge galvannealed steel. The cabinet shall be powder coated with a satin black finish to provide durability, and to protect from corrosion. Armaflex elastomeric thermal insulation shall be used to insulate the cabinet, dampen noise, and prevent damage from vibration. Casters and leveling feet shall be included to ease the installation and level the equipment with existing IT solutions. gauge base and the top plate. The top plate, front and rear panels shall be fabricated from 16 gauge galvannealed steel.

The cabinet shall be powder coated to provide durability, and to protect from corrosion. Armaflex elastomeric thermal insulation shall be used to block noise. Casters and leveling feet shall be included to ease the installation and level the equipment with existing IT solutions.

Air Flow Patterns

All units shall be designed using a front discharge with a rear return airflow pattern. An optional plenum with a front diverted discharge pattern is available.

Air Filtration

All units shall be equipped with removable, washable filters. These filters shall consist of an open cell structured polyurethane foam with a roll formed 3000 series aluminum frame. Filters shall meet both UL 900 and UL 94 HF-1 standards.

Mechanical Components

Backward Inclined, Plenum Style Fan, with an EC Motor

The blowers shall be backward inclined plenum style fans with an Electronically Commutated (EC) motor, for maintenance free operation. The motor shall include:

- Integrated electronic control board
- Direct microprocessor control signaling for fan speed control, soft-starting capabilities
- Integrated current limitations

Each fan shall be low noise, low vibration manufactured with an anti-corrosive aluminum impeller. Each fan impeller shall be dynamically and statically balanced in two planes to minimize vibration during operation.

Refrigeration System

All piping and components contained within the refrigeration system shall be rated for use with R-410A refrigerant. Each refrigeration circuit shall include, as a minimum a refrigerant drier/strainer, sight glass with moisture detector, an electronic thermal expansion valve, an evaporator coil, a compressor, a high pressure switch with manual reset, and a low pressure switch with automatic reset.

Fixed Speed Scroll Compressor

The compressor shall be a high efficiency, high reliability and low noise scroll compressor.
The compressor shall be equipped with internal line break motor protectors, an internal pressure relief valve and external vibration mounting isolation.

Variable Speed Compressor (CRS-090)

The speed of the compressor shall be controlled with a variable frequency drive (VFD) to match compressor capacity to the actual thermal load. The VFD shall be capable of providing power for the compressor to operate up to 120rpm. The VFD shall proportionally control the compressor motor speed to maintain a constant supply air temperature.

Evaporator Coil

The evaporator coil shall be constructed of seamless drawn copper tubes, mechanically bonded to tempered aluminum fins (that have an enhanced design for maximum heat transfer) and mounted in a stainless-steel condensate drain pan. The coil shall be designed for a maximum of 500 ft/min face velocity.

Proportional Electronic Expansion Valve
An electronically operated thermostatic expansion valve shall be installed to precisely control the flow of liquid refrigerant into the evaporator coil while maintaining the desired superheat across a wide range of operating conditions.

Electronic Hot Gas Bypass (CRS-042/084)
An electronically operated hot gas bypass valve shall be installed. The hot gas bypass valve shall provide modulation of the unit's cooling capacity, and evaporator coil freeze protection under low load conditions.

Piping Configuration
Top Piping: When top piping is specified, the CyberRow units shall be provided with connections for water/glycol or refrigerant piping and condensate discharge on the top of the cabinet.
Bottom Piping: When bottom piping is specified (raised floor applications, for example), the CyberRow units shall be provided with connections for water/glycol or refrigerant piping and condensate pump discharges through the bottom of the cabinet.

Condensate Pump (Optional)
The CyberRow shall include an optional factory wired and installed condensate pump. The optional condensate pump shall have the capacity of 13.5 gal/h at 26 ft of lift. The maximum feet of head is 26 ft. The condensate pump shall be piped with either top or bottom discharge connections, to remain consistent with top or bottom piping connections.

Return Air Sensor
A factory mounted and wired temperature sensor(NTC) mounted in the return air stream temperatures shall be provided.

Supply Air Control (Optional)
The STULZ CyberRow shall be provided with a temperature and humidity sensor factory unit mounted in the return air stream and a field installed supply air temperature (optional) and humidity sensor for supply air temp control capabilities. The controller shall provide the user an adjustable supply air control setpoint.

Air Cooled Heat Rejection

-20°F Variable Fan Speed Control (AR)
The air-cooled system shall incorporate a low ambient, variable speed fan, head pressure control. The pressure control shall be for year-round air conditioning system operation down to -20°F DB minimum ambient air temperature.

-30°F Flooded Control (AR)
The air-cooled system shall incorporate a low ambient, flooded head, pressure control. The pressure control shall be for year-round system operation down to -30°F DB minimum ambient air temperature. Liquid refrigerant receivers, with receiver liquid- level sight glass and head pressure regulator valves (for flooded condenser operation) shall be included, but not factory installed

Water/Glycol Cooled Heat Rejection

Stainless Steel Brazed-Plated (W/G)
The evaporator refrigerant circuit shall be provided with a factory installed single pass, counterflow configured, brazed plate heat exchanger, with integral subcooler. It shall be constructed of type 316 stainless steel; designed and tested for a 650 psig w.w.p.

Two-Way, 600 psig Regulating Valve (W/G) (Standard)
The refrigerant circuit head pressure shall be controlled by a factory installed two-way water/glycol regulating valve rated for 400 psig w.w.p. The two-way condenser water modulating valve shall automatically meter the flow of water to the condenser. It shall do so, in response to a proportional signal (0–10 VDC) provided to the valve by the E² controller.

Three-Way, 600 psig Regulating Valve (W/G) (Optional)
The refrigerant circuit head pressure shall be controlled by a factory installed three-way water/glycol regulating valve rated for 400 psig w.w.p. The three-way condenser water modulating valve shall automatically meter the flow of water/glycol to the condenser. It shall do so, in response to a proportional signal (0–10 VDC) provided to the valve by the E² controller.

Electrical System
The electrical system shall conform to National Electrical Code (NEC) requirements. In accordance with NEC Class II requirements, the control circuit shall be 24 volts AC wire, and shall not be smaller than 18 AWG. All wiring shall be neatly wrapped on run in conduit, or cable trays, and routed in bundles. Each wire shall end with a service loop and be securely fastened by an approved method. Each wire in the unit shall be numbered for ease of service tracing.

All electrically actuated components shall be easily accessible from the front of the unit without reaching over exposed high voltage components or rotating parts. Each high voltage circuit shall be individually protected by circuit breakers, or manual motor starters, on all three phases. The blower motor shall have thermal and short circuit protection. Line voltage and 24 volt control circuit wiring shall be routed in separate bundles.
The electric box shall include all the contactors, starters, fuses, circuit breakers, terminal boards, and transformers required for operation of the CyberRow unit. It shall also allow for full service via front and rear access panels.

Remote Stop/Start Contacts
Included in the system's electrical control circuit shall be a 2-pin terminal connection for remote stop/start of the CyberRow air conditioner by remote source.

Main Power Switch Service Switch
The CyberRow unit shall be provided with a unit mounted main power service switch.

E^2 Microprocessor Controller

The advanced microprocessor based E^2 controller shall be equipped with flexible software capable of meeting the specific needs of the application. The setpoints shall be default and their ranges shall be easily viewed and adjusted from the user interface display.
The program and operating parameters shall be permanently stored on a non-volatile system in the event of power failure. The controller shall be designed to manage temperature to a user defined setpoint via control output signals to the CyberRow unit.
The controller shall receive inputs for measurable control conditions (temperature, relative humidity, and dewpoint) via return air or remote mounted supply air sensors. The internal logic will then determine if the conditions require cooling. Control setpoints shall be established to maintain design conditions of the installation. The controller will respond accordingly to changes in these conditions and control the output/demand for the appropriate mode of operation until user defined conditions are achieved.

Field Configurable

The program for the E^2 controller shall be field configurable, allowing the operator the capability of selecting control setpoints specific to the application. Operator interface for the E^2 controller is provided via a door mounted user interface display panel. The display panel shall have a backlit LCD graphical display and function keys giving the user complete control and monitoring capability of the precision cooling system. The menu driven interface shall provide users the ability to scroll through and enter various menu screens.

Password Protection

Access to the Info Menu, Alarms Log, and the ability to monitor room conditions shall be allowed without the use of a password. Modifications to the control setpoints shall require the use of a password. The controller shall be programmed to recognize predetermined security levels before allowing access to display screens containing critical variables. Three secured menu levels (Control, Service, and Factory) will support unique passwords that must be entered to access the menu screens so only authorized personnel may perform modifications to the settings.

Restorable Parameters / Factory Defaults

Upon initial start-up the CyberRow system shall operate using the setpoints programmed by the factory. The customer may enter new operating parameters in the Control menu and the system will then operate accordingly. The new setpoints may be stored as, Customer Default Setpoints. The primary setpoints entered by the factory still remain stored in the controller‘s memory as, Factory Setpoints.
The setpoints for the system may be readjusted in the Control menu at any time. If it becomes necessary, the customer may restore the setpoints back to the Customer Default setpoint values or to the original factory (primary) set point values.

A/C Grouping pLAN Operation

A/C Grouping pLAN Operation

Multiple CyberRow system controllers shall be able to connect (grouped) to a pLAN local network, allowing the communication of data and information from each controller to a central control terminal or lead controller. The lead controller display screens can be used to monitor and adjust group control variables for the individual system controllers. Each E^2 controller connected to the pLAN network shall be identified with its own unique address.

Multiple CyberRow units consisting of up to eight STULZ precision air conditioners equipped with like controllers may be controlled and monitored via the E^2 Microprocessor Controller. With multiple CyberRow units each unit can selectively be configured as Active to operate as a primary A/C, Capacity Assist for staged operation, or as Standby to come online in case of a failed air conditioning unit to ensure continuous availability. The controller may also be configured to rotate units with timed duty cycling to promote equal run-time and ensure that each CyberRow unit within the rotating group is operationally exercised on a periodic timed basis.

BMS Interface (Optional)

The E^2 controller may incorporate a 10 Mbps communication interface port that can be field connected through a serial interface to a Building Management System via Modbus, BACnet, SNMP, or HTTP as configured by the factory. A controller interfaced to a network must be configured for BMS communication.

Alarms, Dry Contact

Alarm conditions shall activate a red LED indicator that backlights the alarm function key. As an option, an alarm condition may also be enunciated by an audible alarm signal. An alarm is acknowledged by pressing the alarm key. This calls up alarm display screens that provide a text message detailing the alarm conditions. After an alarm condition is corrected, the alarm can be cleared by pressing the alarm key.

User Interface Display Panel

The user interface display panel features an easy to read, backlit liquid-crystal alphanumeric display equipped with contrast adjustment and LED illuminated function keys. The screens that appear on the user interface display panel present data that originates from the controller I/O module. The controller is operated via a 6-key menu-driven loop structure and offers an alarm log plus four different interface menu levels to the operator: Information, Control, Service, and Factory. These menus permit the user to easily view, control, and configure operating parameters for the CyberRow unit.

Optional Accessories

Smoke Detection

A photo-electric smoke detector shall be factory installed and wired in the evaporator section of the suction's ide of the evaporator blower. The air conditioner will shut down upon sensing smoke in the return air stream.

Firestat
The air conditioner shall be provided with a factory wired and mounted firestat. The firestat will shut down the air conditioner upon sensing a high return air temperature.

Remote Water Detector Spot Type
A remote single point water and leak detector shall be factory supplied and shall ship separately for field installation. Upon sensing a water leak, the normally closed water detector control circuit shall open, thereby

Remote Water Detector Strip Type
A 20 ft long remote strip/cable type water and leak detector shall be provided for remote field installation. Upon sensing a water leak, the normally closed water detector control circuit shall open, thereby shutting down the CyberRow unit's water producing components.

Code Conformance

The supplied system meets the following compliance approvals:
- $_{C}$**ETL**$_{US}$ listed and conforms to UL standard 60335-1 and UL standard 60335-2-40
- Certified to CSA standard C22.2 # 60335-1 and CSA standard C22.2 # 60335-2-40


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